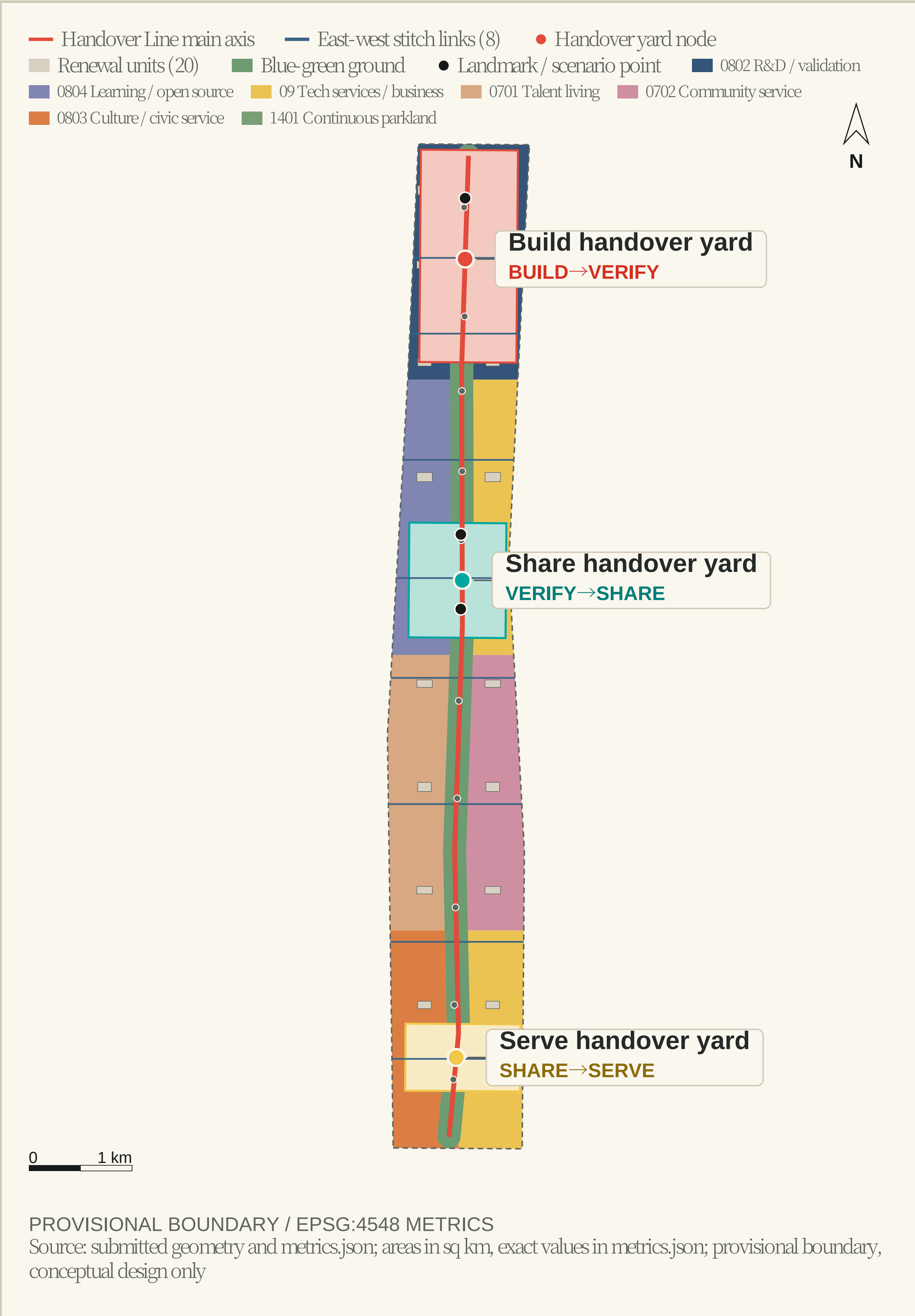


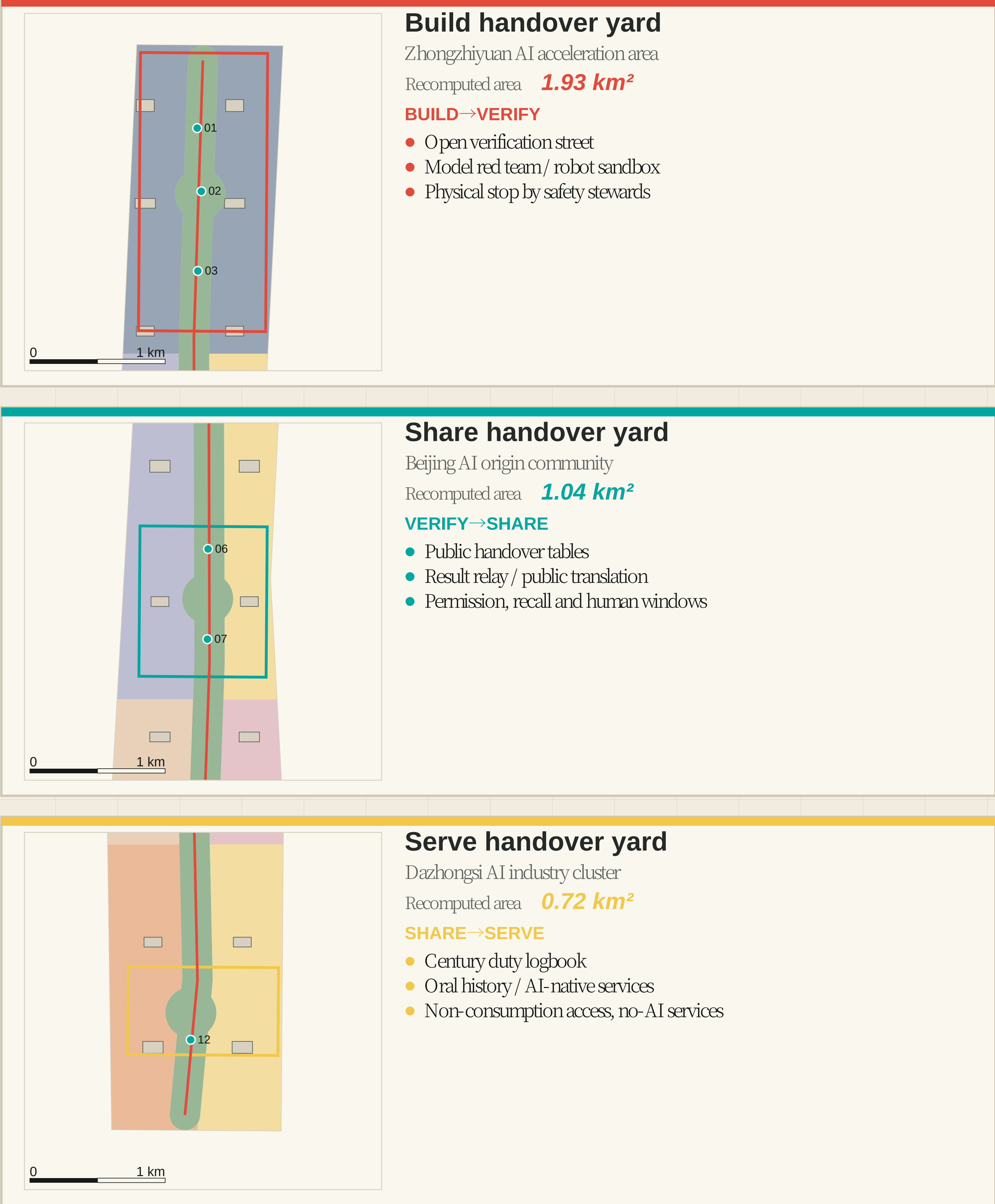
# JING-ZHANG HANDOVER LINE

From railway shift handover to a launchable, reviewable and reversible civic-AI pilot

SITE OVERVIEW



THREE HANDOVER YARDS



HANDOVER PROTOCOL

- 01 / BUILD TO VERIFY

Models, robots, compute and data enter tests with versions, risks and shutdown rules.
- 03 / SHARE TO SERVE

Civic services retain a human desk, a no-AI equivalent, appeal and exit records.

- 02 / VERIFY TO SHARE

Results, licences, failures and limits travel together; open source never means exposing personal data.
- 04 / RETURN THE PROBLEM

Complaints, repairs and shutdown causes travel back to research as public learning.

FIRST PILOT / P0 PUBLIC HANDOVER TABLE

The overall concept compressed into one launchable, reviewable and reversible S-class delivery unit

|                                                 |                                                                      |                                                                     |                                                                    |
|-------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|
| <div>12/12</div> <div>OFFLINE TASKS</div>       | <div>C1</div> <div>STAFFED DESK</div> <div>Before smart UI</div>     | <div>C2</div> <div>NO-AI EQUAL</div> <div>Same outcome/access</div> | <div>C3</div> <div>VISIBLE STOP</div> <div>No app or sign-up</div> |
| <div>48/48</div> <div>TAKEOVER ASSERTIONS</div> | <div>C4</div> <div>APPEAL + EXIT</div> <div>One place to begin</div> | <div>C5</div> <div>DUAL LOGBOOK</div> <div>Giver + receiver</div>   | <div>C6</div> <div>ACCESS ROUTE</div> <div>Continuous, clear</div> |
| <div>0/12</div> <div>FIELD TASKS</div>          |                                                                      |                                                                     |                                                                    |

The tabletop rehearsal is not presented as field evidence

SPACE-TO-ROLLBACK PROOF CHAIN

SPACE → USER → ROLE PAIR → COST S → SIX GATES → 12-TASK OFFLINE REHEARSAL → HUMAN ROLLBACK

COST S = removable equipment E + maintenance M + training T + independent verification V + 10% contingency

SITE | FIRE | ACCESS | PRIVACY | MAINTENANCE | PUBLIC NOTICE: ALL SIX BEFORE FIELD USE

12/12 and 48/48 prove only a complete synthetic offline takeover chain; field safety, accessibility, public acceptance and approval remain pending.

PHASE GATES

Gate-driven progression, no calendar commitment

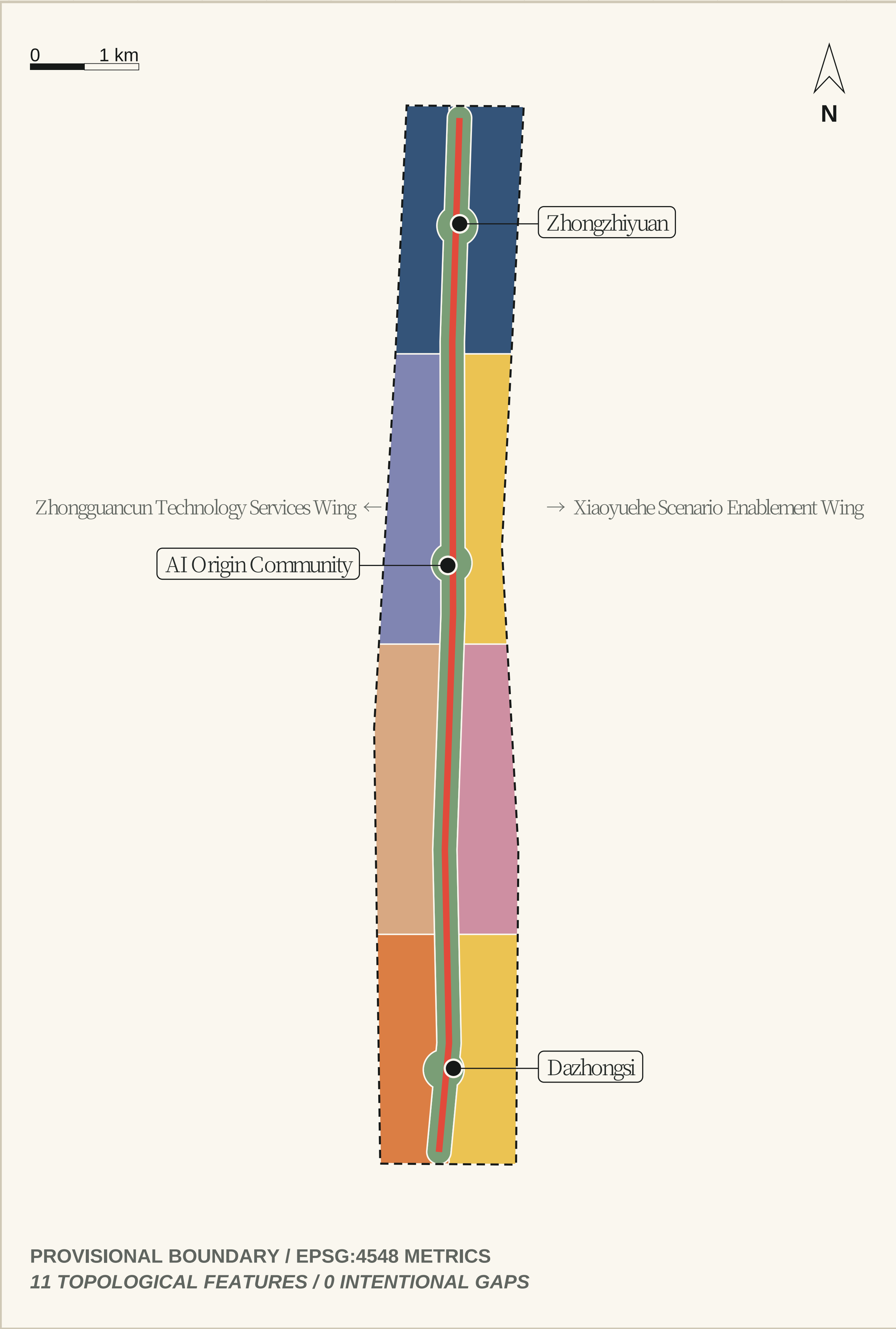
|                                                                                                                                               |                                                                                                                                             |                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <div>PHASE 1 / PHASE-001</div> <div>Baseline and southern civic service pilot</div> <div>Recomputed area</div> <div>4.84 km<sup>2</sup></div> | <div>PHASE 2 / PHASE-002</div> <div>Share handover yard and the middle link</div> <div>Recomputed area</div> <div>3.55 km<sup>2</sup></div> | <div>PHASE 3 / PHASE-003</div> <div>Build handover yard and belt operations</div> <div>Recomputed area</div> <div>3.03 km<sup>2</sup></div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|



# Three Scopes, Land-use Topology and Key Areas

Full coverage without statutory fiction; three yards, three shifts of duty

## Land-use topology and three scopes



### LAND USE

- 0802 R&D / validation
- 0804 Learning / open source
- 09 Tech services / business
- 0701 Talent living
- 0702 Community service
- 0803 Culture / civic service
- 1401 Continuous parkland
- Handover Line main axis
- Handover yard node

**43.6 km<sup>2</sup>** → **11.4 km<sup>2</sup>** → **3 YARDS**  
RESEARCH Industry + future city      OVERALL DESIGN Spatial + delivery system      KEY AREAS Detailed validation

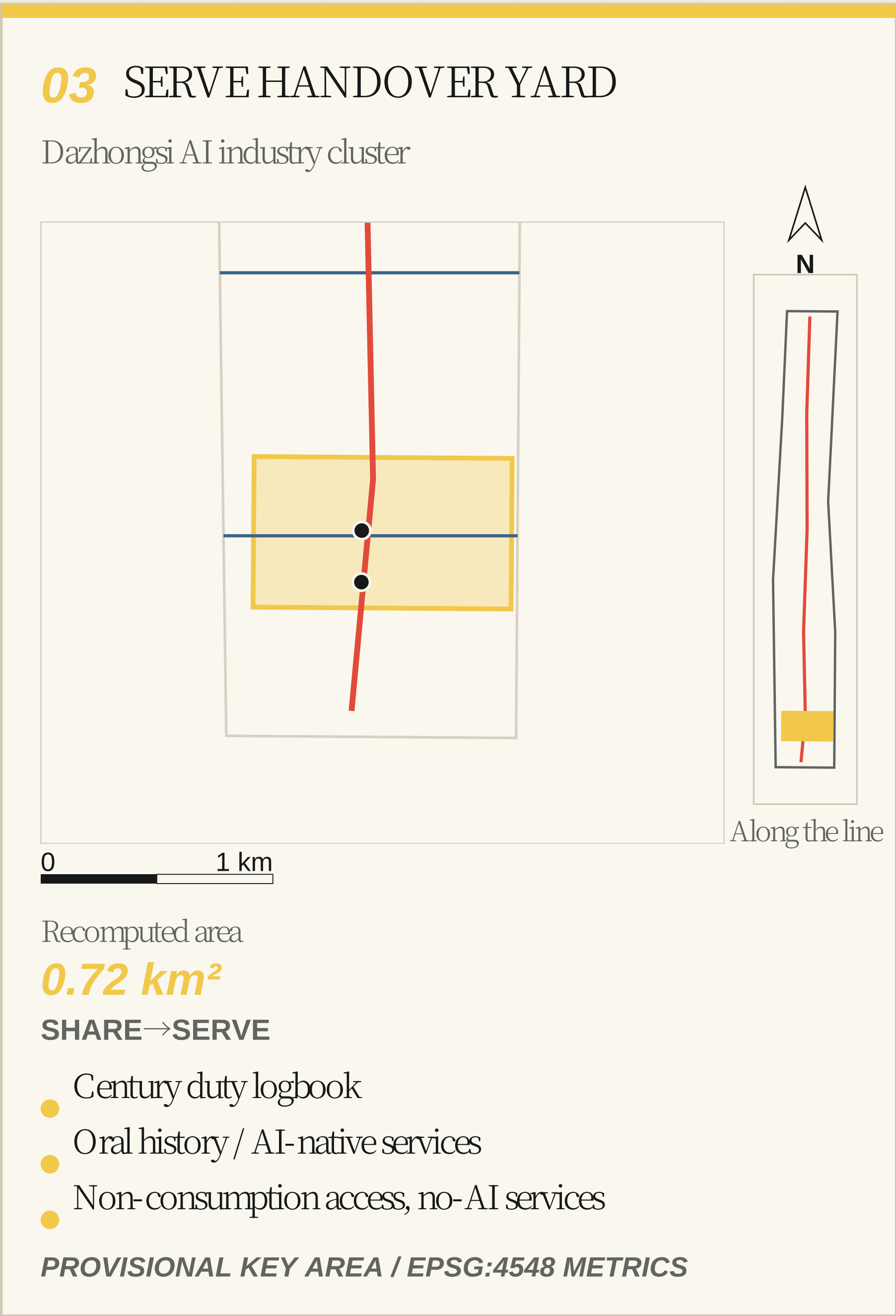
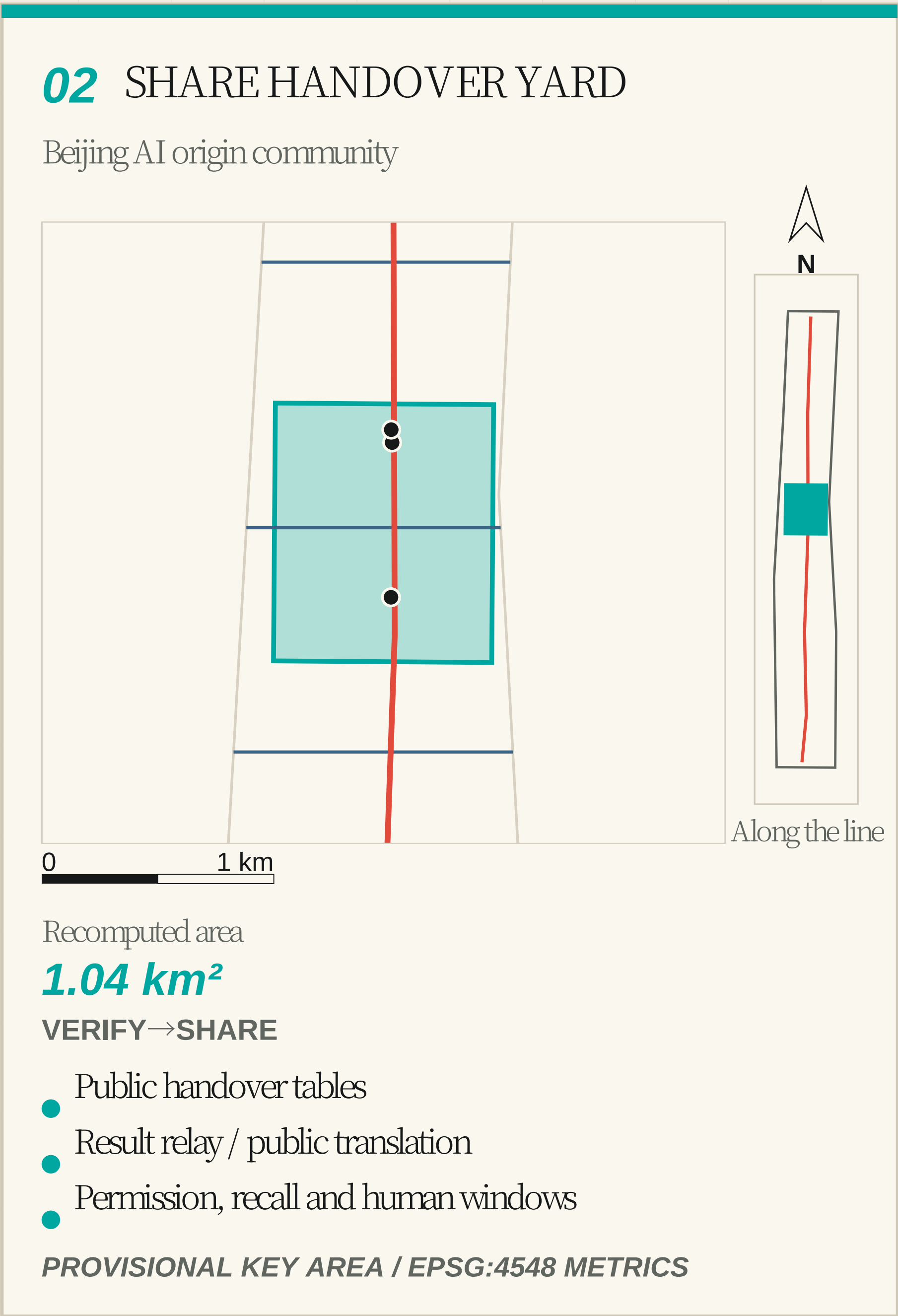
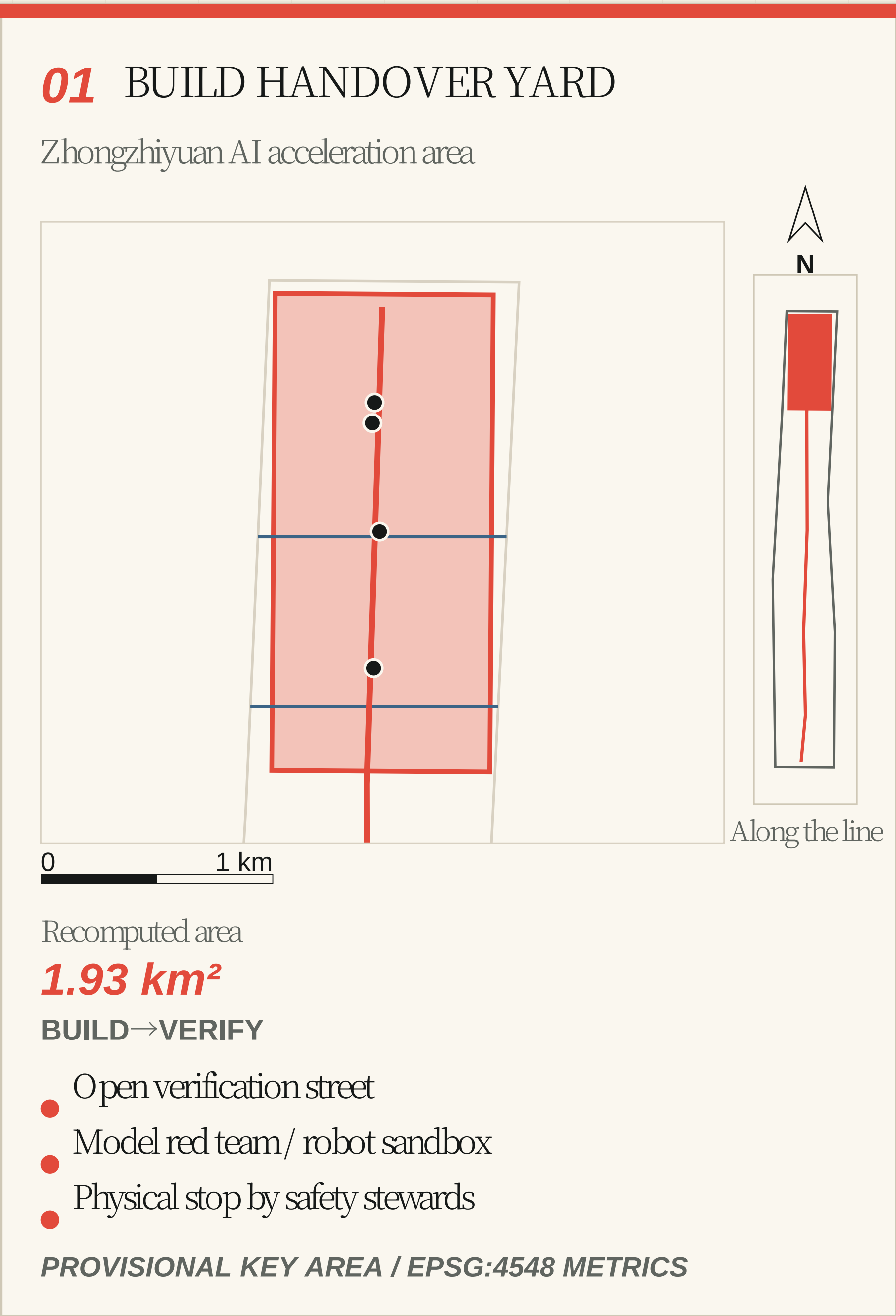
Design recommendation: all zones recalculate when official boundaries arrive; no ownership or statutory land-use claim.

Data: land\_use.geojson and site\_boundary.geojson; codes from the public MNR subset

## Three handover yards, three shifts of duty

Each key area answers programme, space, scenario and human-takeover gates  
Extents from key\_areas.geojson; points from public\_space.geojson; one shared scale

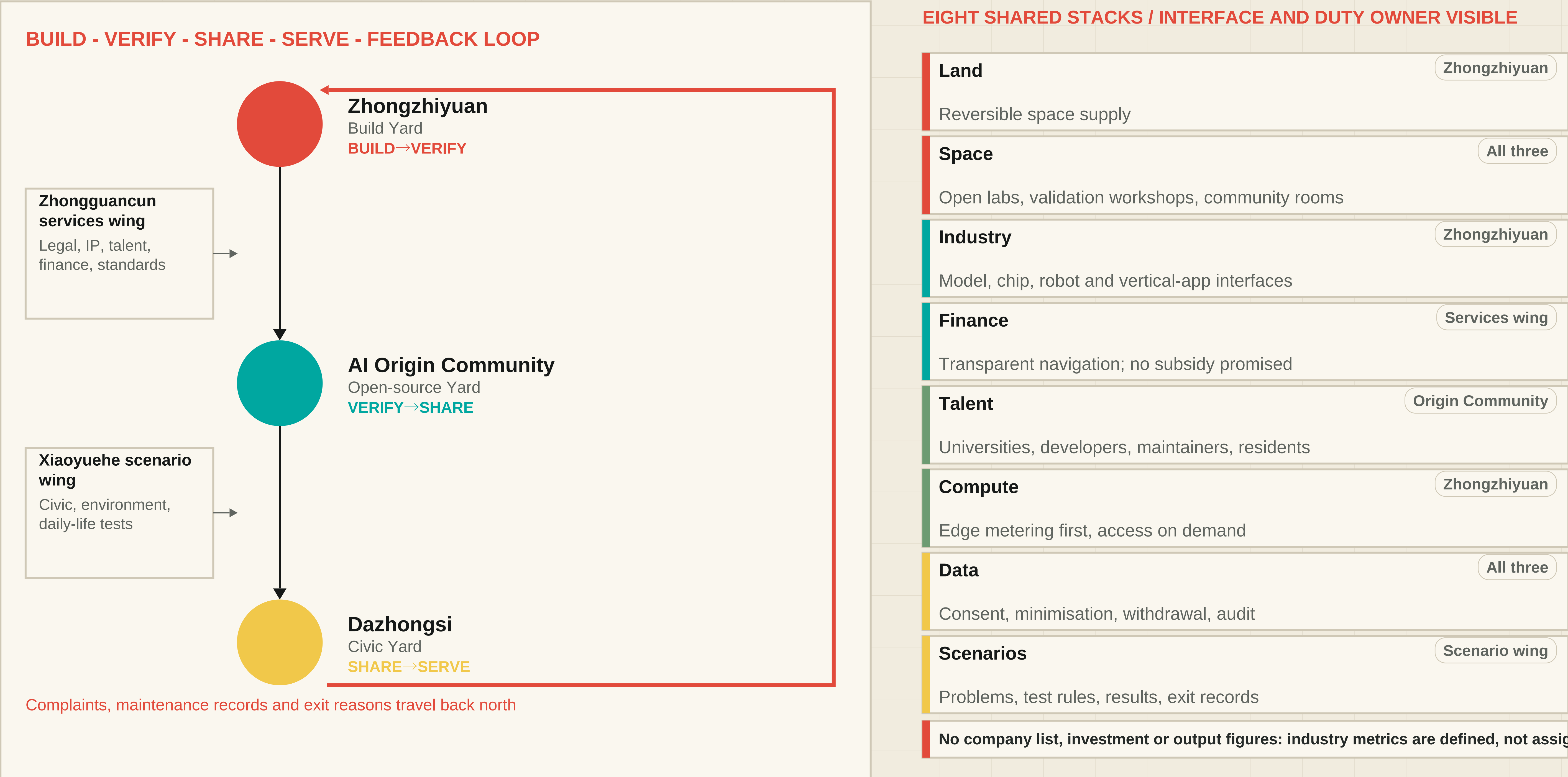
- Handover Line main axis
- East-west stitch links
- Public scenario / landmark





# Ecosystem Map and Regional Innovation Synergy

Eight shared stacks x three areas, two wings; synergy across belt-city-municipal-regional tiers



|             |                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WITHIN BELT | Three areas and two wings organise the research-validation-service loop; objections and shutdown reasons travel back north.                                                    |
| WITHIN CITY | Connects to the official “two zones and one belt” framework, matching Zhongguancun technology supply to this belt's validation and scenario demand.                            |
| MUNICIPAL   | Future Science City, Huairou Science City and BDA form an original-research / engineering-validation / scale-delivery division; this belt is the public-validation output end. |
| REGIONAL    | Compute, energy and scenario interfaces reserved toward Zhangjiakou and other Beijing-Tianjin-Hebei nodes; feasibility to be confirmed by authorities.                         |

Eight stacks x three areas / two wings — placement matrix

| Shared stack | Zhongzhiyuan | Origin Community | Dazhongsi   | Services wing | Scenario wing |
|--------------|--------------|------------------|-------------|---------------|---------------|
| Land         | <div></div>  |                  |             |               |               |
| Space        | <div></div>  | <div></div>      | <div></div> |               |               |
| Industry     | <div></div>  |                  |             |               |               |
| Finance      |              |                  |             | <div></div>   |               |
| Talent       |              | <div></div>      |             |               |               |
| Compute      | <div></div>  |                  |             |               |               |
| Data         | <div></div>  | <div></div>      | <div></div> |               |               |
| Scenarios    |              |                  |             |               | <div></div>   |

Placement follows the duty-owner tags at upper right; “All three” = one dot each in Zhongzhiyuan, Origin Community and Dazhongsi, not a separate fourth owner

Phase gates (three sequential gates, no calendar commitment)

|                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>PHASE 1 / PHASE-001</b> 4.84 km²</p> <p><b>Baseline and southern civic service pilot</b></p> <p><b>Entry condition</b></p> <p>Official boundary and key-area extents available; once condition, ownership, user and accessibility baselines are complete, low-risk human-led pilots may begin.</p> <p>Rollback: withdraw the temporary intelligent layer; keep fixed wayfinding, lighting, access and human service.</p> | <p><b>PHASE 2 / PHASE-002</b> 3.55 km²</p> <p><b>Share handover yard and the middle link</b></p> <p><b>Entry condition</b></p> <p>Inherit Gate 1's outcomes; once ownership, fire safety, community consultation, accessibility and ongoing operating duties clear review, first-floor opening and result relay may proceed.</p> <p>Rollback: revert to Gate 1 usable state; halt expansion and restore ordinary ground floor and community human service.</p> | <p><b>PHASE 3 / PHASE-003</b> 3.03 km²</p> <p><b>Build handover yard and belt operations</b></p> <p><b>Entry condition</b></p> <p>Inherit Gate 2's outcomes; once transport, utilities, heritage, privacy, safety, emergency response and ongoing budget clear professional review, cross-area validation networks may be discussed.</p> <p>Rollback: retract to validated segments; halt cross-area linkage and keep basic civic service and heritage interpretation.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



# Renewal structure and building decisions

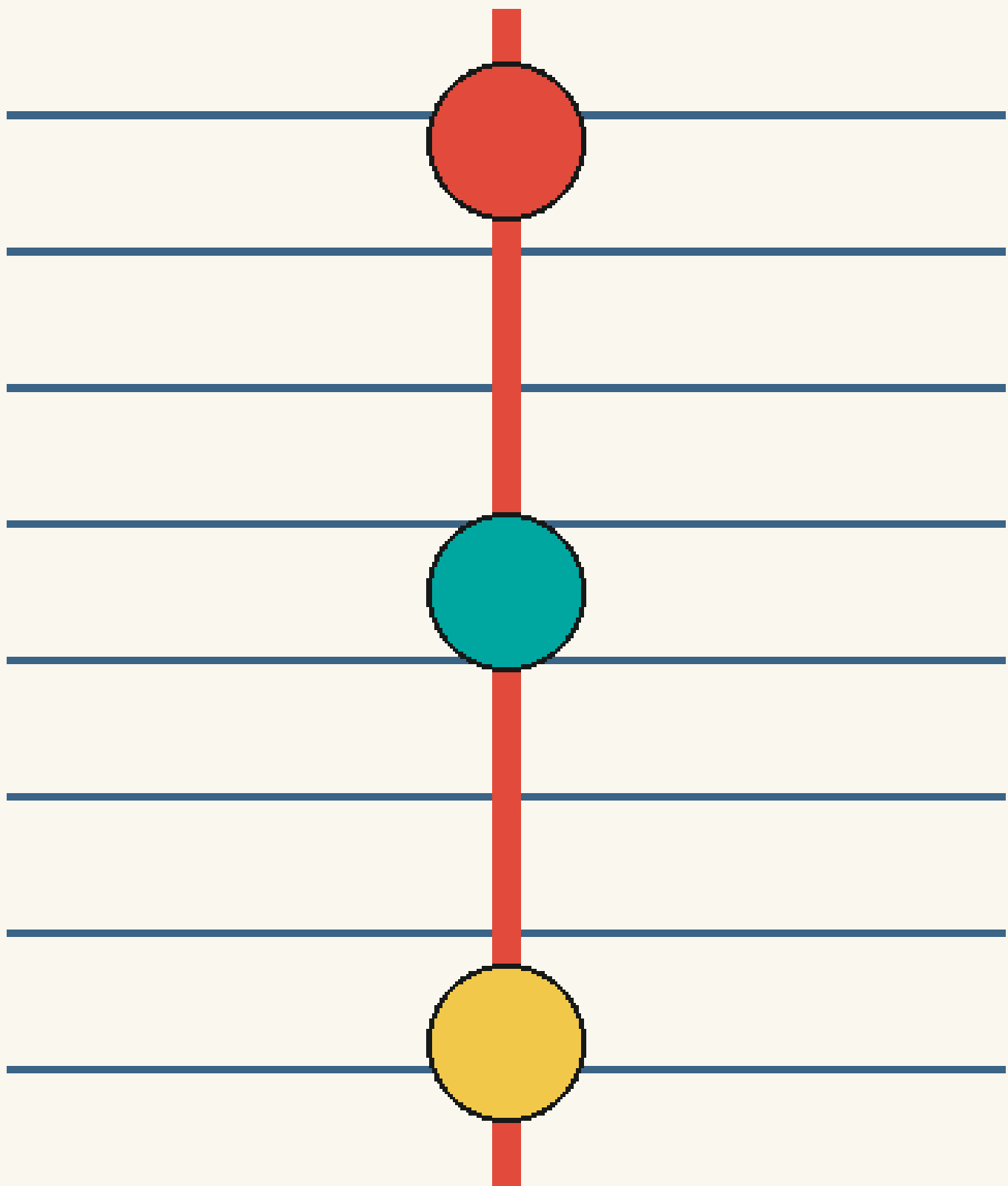
One spine, eight stitches, three yards; priority areas in three tiers, buildings in four steps

F / 07

## Renewal structure and priority renewal areas

One spine, eight stitches, three yards; three tiers, operate first, reversible next, fixed last

### Renewal spatial structure



Handover Line / renewal spine  
East-west stitch links ×8  
Handover yards ×3

### Priority renewal tiers

#### Tier 1

The three handover yards and their walkable surrounds  
Led by a coordinating body, multiple owners join by agreement

#### Tier 2

Stitch-spine junctions and low-efficiency space beside the heritage park  
Renewal units bundled by district, one public interface

#### Tier 3

The remaining frontage along the line  
Signage, shade and frontage tidying only; ownership untouched

### Twenty typology test units (10 per side)

Verify workshop  
5  
26.6%

Shared service stack  
5  
26.6%

Open R&D house  
5  
23.4%

Community co-op house  
5  
23.4%

Retain-first adaptive reuse 14 • investigate building by building 6

Source: submitted geometry buildings/roads/key\_areas/phasing; conceptual structure, no demolish-or-keep conclusion

JING-ZHANG HANDOVER LINE / PACKAGE v1.15

R / RETAIN

### Keep in use

Keep in use and complete the condition record.

A / ADAPT

### Low-impact repair

Fire safety, energy, accessibility and open ground floor first.

I / INSERT

### Removable trial

Removable elements and short-term use only.

D / INVESTIGATE

### Not demolition

Awaits survey, structure, ownership, fire and heritage review.



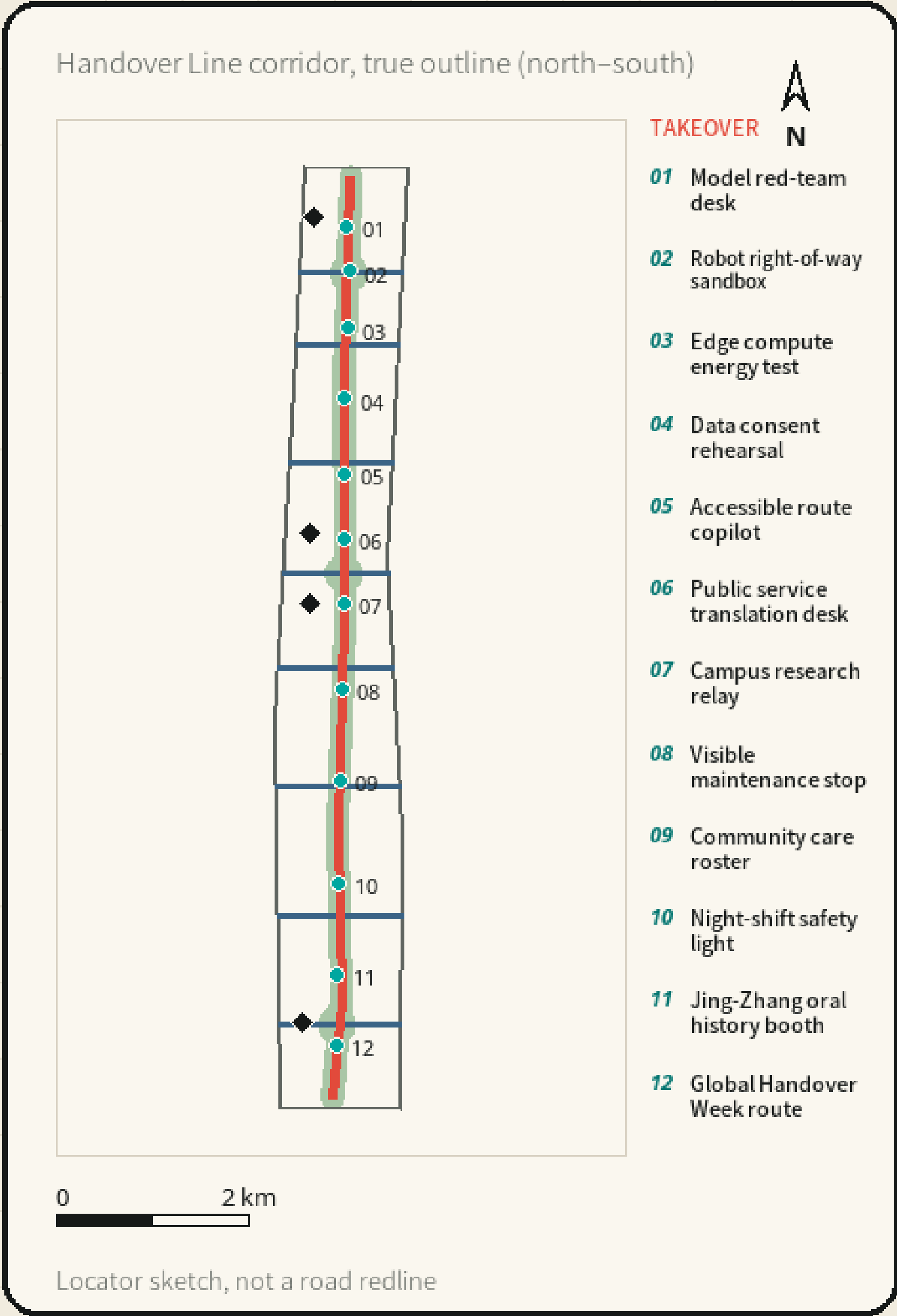
# Twelve reversible scenarios and human takeover

Movement, blue-green ground, human takeover and no-AI equivalents

F / 04

## One spine, eight stitch links, twelve takeover points

The smart layer is removable; basic access and human service stay available



### MOVEMENT / SERVICE

- Continuous walk-cycle spine
- East-west stitch links (8)
- Blue-green comfort ground
- Human-takeover scenario node (SCN)
- Civic landmark point

### A OFFLINE BY DEFAULT

Static signs, light, tactile maps and human help do not rely on a model.

### B GRADED TAKEOVER

Low confidence, boundary breach, objection or outage calls the duty person.

### C EXIT IS RECORDED

Shutdown cause, deletion and service continuity enter the public logbook.

### SECTION The handover section: order and adjacency only, no dimensions

| Handover counter                                                                                                                                  | Accessible walk                                                                     | Takeover Kiosk Zero                                                                        | Retained track                                           | Machine-only lane                                                                       | Planting and rest                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Walking surface (continuous, no level change)<br>The staffed counter precedes the smart interface and still works once the smart layer is removed | Never interrupted by equipment, advertising, temporary occupation or charging posts | The physical stop control is reachable from the footway without downloading or registering | Set flush in the paving, a trace rather than an obstacle | Separated from walking; crossing the edge stops the machine, safety staff take priority | Non-commercial dwelling; seats and shade carry no spending condition |

Adjacency rule: counter, walk and stop control stay continuously reachable; the machine lane never comes between them

Conceptual connectivity, not road redlines or engineering feasibility; lengths calculated in EPSG:4548

JING-ZHANG HANDOVER LINE / PACKAGE v1.15

SCN-01

### Model red-team desk

Human review + paper risk log

SCN-02

### Robot right-of-way sandbox

Safety staff + physical stop

SCN-03

### Edge energy test station

Manual shutdown + offline meter

SCN-04

### Data-consent rehearsal

Paper consent + deletion desk

SCN-05

### Accessible-route copilot

Tactile map + human direction

SCN-06

### Public translation desk

Human desk + multilingual paper

SCN-07

### Campus research relay

Human curation + open catalogue

SCN-08

### Visible maintenance stop

Phone report + human dispatch

SCN-09

### Community care roster

Staff telephone roster

SCN-10

### Night-shift safety light

Fixed light + human patrol

SCN-11

### Jing-Zhang oral-history booth

Human guide + paper catalogue

SCN-12

### Global Handover Week route

Fixed signs + volunteers



# Recalculable, exitable, continuable

Metrics, phasing, risk, sources and professional review gates

F / 05

## The first pilot already has a reviewable proof chain

PUBLIC HANDOVER TABLE P0 | 12/12 OFFLINE TASKS | 48/48 TAKEOVER ASSERTIONS | 0 FIELD TASKS

**PARTICIPANT-CONTROLLED OFFLINE TABLETOP**

12/12

OFFLINE TASKS

48/48

TAKEOVER ASSERTIONS

0/12

FIELD TASKS

The tabletop rehearsal is not presented as field evidence

12 SCENARIOS x 4 TAKEOVER CHECKS

|    |    |    |    |
|----|----|----|----|
| 01 | 02 | 03 | 04 |
| 05 | 06 | 07 | 08 |
| 09 | 10 | 11 | 12 |

SMART LAYER OFF HUMAN TAKEOVER SYNTHETIC DATA CLEARED HUMAN SERVICE RESTORE

### P0 PUBLIC HANDOVER TABLE / FIRST PILOT UNIT

Six components form one launchable, removable and auditable S-class pilot package

C1

STAFFED DESK

Before smart UI

C2

NO-AI EQUAL

Same outcome/access

C3

VISIBLE STOP

No app or sign-up

C4

APPEAL + EXIT

One place to begin

C5

DUAL LOGBOOK

Giver + receiver

C6

ACCESS ROUTE

Continuous, clear

COST S = removable equipment E + maintenance M + training T + independent verification V + 10% contingency

SIX DUTY TYPES: space operator | community service | accessibility | privacy/data | independent review | public-realm maintenance

01

SPACE

P0 TABLE

02

USER

RESIDENT/VISITOR

03

ROLE PAIR

GIVER+RECEIVER

04

COST

CLASS S

05

GATES

ALL SIX

06

REHEARSAL

12/12

07

ROLLBACK

HUMAN SERVICE

PROVES: readable structure, auditable ledger and a complete offline takeover route.  
DOES NOT PROVE: field safety, measured accessibility, public acceptance or authority approval.

simulation.json × metrics.json × assumptions.json; synthetic offline tabletop, no live service or personal data

JING-ZHANG HANDOVER LINE / PACKAGE v1.15

PHASE 01

Return to baseline: official data, inventory, wayfinding and three low-risk human-led pilots.

PHASE 02

Open-source yard: ground floors, research relay and community service after ownership and community review.

PHASE 03

Belt-wide operation: complex validation networks only after transport, utilities, heritage, privacy and safety conditions.

Unknown by design: FAR, height, ownership, road redlines, utility capacity, investment and output.